

MATH 3430 - Week 5

Nonlinear Equations and Systems

1 Overview

Much of our interest in this course is with systems of first-order differential equations:

$$\begin{aligned}x' &= f(x, y) \\ y' &= g(x, y)\end{aligned}$$

We encountered these systems during week 2, when we focused primarily on **linear systems** with $f(x, y) = ax + by$ and $g(x, y) = cx + dy$. This week we will look at **nonlinear systems** where f and g are no longer restricted to be linear functions of x and y .

Our first objective is to show how the behavior of a nonlinear system near an equilibrium point can be understood by constructing a related linear system, called the **linearization** of the original nonlinear system. This linear system is in some sense the “best” linear approximation to the nonlinear system. This can be thought of as a process very similar to using a tangent line (the “best” linear approximation) to understand the behavior of a curve (which is usually nonlinear).

Our next objective is to consider certain families of nonlinear systems that have enough additional structure to allow us to construct either explicit solutions or gain a qualitative understanding of what’s going on (at least near equilibrium points). These families include **Hamiltonian Systems**, **Dissipative Systems** and **Gradient Systems**.

Finally, we will look at nonlinear systems that exhibit what are called **limit cycles**. We will focus primarily on the **Van der Pol Equation** and will explore some of its behavior.

While we will be able to develop some analytical tools, the approach this week will be primarily qualitative and numerical. We will use Sage extensively, but at this stage in the semester this shouldn’t pose too many problems.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) Nonlinear Systems. The Pendulum. Linearization.
- (T) Hamiltonian Systems. Energy.
- (W) Predator-Prey Systems. Dissipative Systems. Lyapunov Functions.
- (R) Wrap-Up and Review
- (F) **In Class Exam!**

Here is a checklist of things to do this week. This is meant to help you keep track of what you've completed and what you still need to work on.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Judson and Lebl.
- ☐ Turn in the Take Home on Wednesday.
- ☐ Complete all of the homework exercises.
- ☐ Make a reference sheet for the in class exam.

3 Terminology and Notation

- Linearization
- Hamiltonian System: a system is **Hamiltonian** if there exists a function $H(x, y)$ such that

$$\begin{aligned}\frac{dx}{dt} &= \frac{\partial H}{\partial y}(x, y) \\ \frac{dy}{dt} &= -\frac{\partial H}{\partial x}(x, y)\end{aligned}$$

- Energy Function (Hamiltonian)
- Gradient System: a system is a **Gradient System** if there exists a function $S(x, y)$ such that

$$\begin{aligned}\frac{dx}{dt} &= \frac{\partial S}{\partial x}(x, y) \\ \frac{dy}{dt} &= \frac{\partial S}{\partial y}(x, y)\end{aligned}$$

- Lyapunov Function
- Limit Cycle

4 Equations and Models

- Pendulum
 - Undamped/Ideal Pendulum

Consider an object of mass m attached to a fixed rod of length L (assume the mass of the rod is negligible). Let g be the force due to gravity. Let θ be the angle that the rod makes in the counterclockwise direction from the negative vertical axis. See the following diagram.

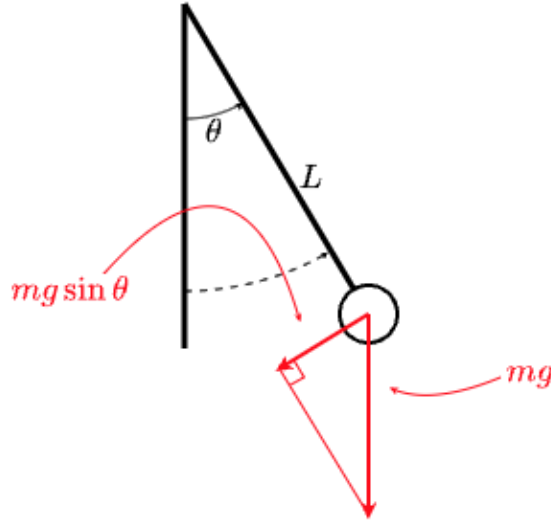


Figure 1: Diagram of a Nonlinear Pendulum

Let $\theta(t)$ give the position of the object at time t and assume that at $t = 0$ the pendulum is hanging in a downward vertical position. If we make the simplifying assumption that the only force acting on the pendulum is gravity, then by Newton's Second Law we have that

$$mL \frac{d^2\theta}{dt^2} = -mg \sin \theta$$

If we introduce the new variable $v = \theta'$ then this second-order differential equation can be written as the linear system

$$\begin{aligned} \theta' &= v \\ v' &= -\frac{g}{L} \sin \theta \end{aligned}$$

– Damped Pendulum

If we also allow air-resistance, then we get a damping term that is proportional to the velocity (and acts in the opposite direction). Let $b \geq 0$ be the damping constant. Then Newton's Second Law gives us

$$mL \frac{d^2\theta}{dt^2} = -bL \frac{d\theta}{dt} - mg \sin \theta$$

If we introduce the new variable $v = \theta'$ then this second-order differential equation can be written as the linear system

$$\begin{aligned} \theta' &= v \\ v' &= -\frac{b}{m}v - \frac{g}{L} \sin \theta \end{aligned}$$

If $b = 0$ then this reduces to the ideal pendulum considered above.

- Van der Pol Oscillator

The Van der Pol Oscillator arises when considering certain electrical circuits. It is the solution to the following differential equation

$$x'' - \mu(1 - x^2)x' + x = 0$$

where $\mu > 0$ is a positive constant.

5 Important Formulas (analytical material)

- The Jacobian matrix of the system

$$\begin{aligned}x' &= f(x, y) \\ y' &= g(x, y)\end{aligned}$$

at the equilibrium point (x_0, y_0) is given by

$$J(x_0, y_0) = \begin{pmatrix} f_x(x_0, y_0) & f_y(x_0, y_0) \\ g_x(x_0, y_0) & g_y(x_0, y_0) \end{pmatrix}$$

- The system

$$\begin{aligned}x' &= f(x, y) \\ y' &= g(x, y)\end{aligned}$$

is Hamiltonian if

$$\frac{\partial f}{\partial x} = -\frac{\partial g}{\partial y}$$

- The Hamiltonian for the undamped harmonic oscillator is given by

$$H(y, v) = \frac{1}{2}mv^2 + \frac{1}{2}ky^2$$

- The Hamiltonian for the ideal pendulum is given by

$$H(\theta, v) = \frac{1}{2}v^2 - \frac{g}{L} \cos \theta$$

- The Jacobian matrix of a Hamiltonian System at an equilibrium point (x_0, y_0) always has the form

$$\begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix}$$

- The eigenvalues of a Hamiltonian System always have the form

$$\lambda = \pm \sqrt{\alpha^2 + \beta\gamma}$$

- A function $L(x, y)$ is a Lyapunov function for a system of differential equations if, for every solution $(x(t), y(t))$ that is not an equilibrium solution of the system,

$$\frac{d}{dt} [L(x(t), y(t))] \leq 0$$

- The system

$$\begin{aligned} x' &= f(x, y) \\ y' &= g(x, y) \end{aligned}$$

is a **Gradient System** if there is a function S such that $\nabla S = (\frac{\partial S}{\partial x}, \frac{\partial S}{\partial y}) = (f(x, y), g(x, y))$.

- To test if a system is a gradient system, it is enough to show that

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$$

- The Jacobian matrix of a Gradient System at an equilibrium point (x_0, y_0) always has the form

$$\begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$$

- The eigenvalues of a Gradient System always have the form

$$\frac{1}{2}(\alpha + \gamma) \pm \frac{1}{2}\sqrt{(\alpha - \gamma)^2 + 4\beta^2}$$

- Because $(\alpha - \gamma)^2 + 4\beta^2 \geq 0$, the eigenvalues of a Gradient System are always real. This means that a Gradient System cannot have equilibrium points that are spiral sinks, spiral sources, or centers.
- The function $-S$ is a Lyapunov function for any Gradient System.

6 Geometric and Visual Aspects (qualitative material)

- The solution curves of a Hamiltonian System lie on the level sets of the Hamiltonian Function.
- Hamiltonian Systems can never have sinks or sources of any type.
- Gradient Systems do not have spiral sinks, spiral sources, or centers (periodic solutions).
- Table summarizing the behavior of the equilibrium points of a linearized system in terms of the eigenvalues of the Jacobian matrix at the equilibrium point.

Eigenvalues of the Jacobian matrix	Behavior	Stability
real and both positive	source / unstable node	unstable
real and both negative	sink / stable node	asymptotically stable
real and opposite signs	saddle	unstable
complex with positive real part	spiral source	unstable
complex with negative real part	spiral sink	asymptotically stable

Figure 2: Behavior of Equilibrium Points of the Linearized Systems

- Limit Cycle (thicker red curve) of the Van der Pol equation.

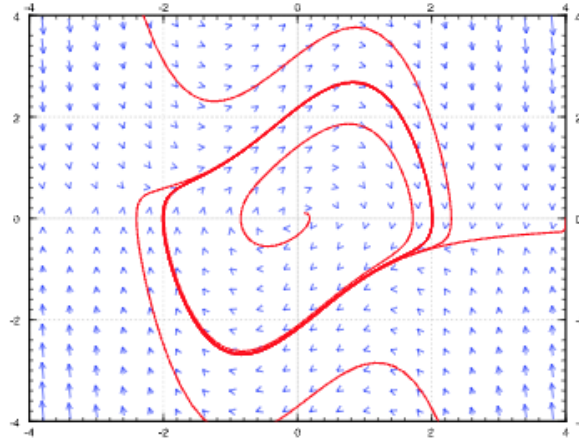


Figure 3: Limit Cycle of the Van der Pol equation

7 Sage Code (numerical material)

7.1 Commands

There aren't any new commands for this material.

7.2 Example Code

- This code generates and plots the (normalized) vector field for the undamped pendulum. It includes three sets of level curves. There is a second plot that shows a solution curve extracted from the `desolve_system_rk4` command:

```
k, M = 0, 1
x, y, t = var('x y t')
F = [y, -k*y - M*sin(x)]
n = sqrt(F[0]^2 + F[1]^2)
p1 = plot_vector_field((F[0]/n, F[1]/n), (x, -4*pi, 4*pi), (y, -4, 4), aspect_ratio = pi)
p2 = plot_vector_field((F[0]/n, F[1]/n), (x, -4*pi, 4*pi), (y, -4, 4), aspect_ratio = pi)
H(x, y) = (1/2)*y**2 - cos(x)
P1 = desolve_system_rk4(F, [x, y], ics=[0, 0, 1.2], ivar = t, end_points = 30, step = 0.1)
S1 = [ [j, k] for i, j, k in P1]
p2 += line(S1, thickness = 2)
p1 += implicit_plot(H, (-4*pi, 4*pi), (-4,4))
p1 += implicit_plot(H - 1, (-4*pi, 4*pi), (-4,4), color = 'red')
p1 += implicit_plot(H - 2, (-4*pi, 4*pi), (-4,4), color = 'green')
p1.show()
p2.show()
```

- This code generates and plots the (normalized) vector field for the damped pendulum, shows one solution curve, and includes the plots of the angle and velocity functions on a separate set of axes:

```
k, M = 0.1, 1
x, y, t = var('x y t')
```

```

F = [y, -k*y - M*sin(x)]
n = sqrt(F[0]^2 + F[1]^2)
p = plot_vector_field((F[0]/n, F[1]/n), (x, -4*pi, 4*pi), (y, -4, 4), aspect_ratio = pi,
axes_labels = ['Angle', 'Velocity'])
H(x, y) = (1/2)*y**2 - cos(x)
P1 = desolve_system_rk4(F, [x, y], ics=[0, 0, 2.5], ivar = t, end_points = 30, step = 0.1)
S1 = [ [j, k] for i, j, k in P1]
p += line(S1)
S2 = [ [i, j] for i, j, k in P1]
S3 = [ [i, k] for i, j, k in P1]
p2 = line(S2, color = 'red', legend_label = 'Angle', legend_color = 'red')
p2 += line(S3, color = 'green', legend_label = 'Velocity', legend_color = 'green')
p.show()
p2.show()

```

8 General Comments

- The unfortunate (or fortunate, depending on your point of view) fact of the matter is that the world is highly nonlinear in nature. The fortunate (and this is undoubtedly good news) fact is that linear approximations can give us a ton of insight and understanding about how this nonlinear world works. But, there are limitations to these linear approximations, and that is just part of the deal.
- Certain types of nonlinear systems have enough regularity that there are explicit things that we can do to understand and work with them. The primary example that we will look at are Hamiltonian systems. Any time that you are given extra structure, you should use it.
- Hamiltonian systems should remind you a lot of what you did at the end of Calc 3. This is not a coincidence. Most of the ideas in Calc 3 were developed in order to solve more complicated differential equations (for what it's worth, most of the ideas in Calc 1 and Calc 2 were developed in order to solve fairly simple differential equations).

9 Reading Notes

- Nonlinear Systems (J 5.1) (L 8.1)
- Linearization (J 5.1) (L 8.1, 8.2)
- Hamiltonian Systems (J 5.2) (L 8.2.4)
- The Pendulum (J 5.2.1) (L 8.3.1)
- Dissipative Systems (J 5.3)
- Gradient Systems (J 5.3)
- Limit Cycles, The Van der Pol Equation (L 8.4)

10 Additional Topics

Look in the indices of Judson and Lebl for these topics.

- The Hopf Bifurcation
- Poincaré Return Map (Poincaré Section)
- Poincaré-Bendixson Theorem

11 Homework Exercises

11.1 To Turn In

Please see the document in the Google Drive for this week's homework problems. Please note that these are due **THURSDAY**, 7/30, at the start of class.